Pair-Bound Lyapunov Stability for a Scalar Dissipative Kuramoto-Type Mean-Field Equation, Formalized in Lean 4

Abstract

We prove global stability ($r(t) \to r^*$ for all initial data with $\alpha(\omega, 0) \in (0, 1)$) for a scalar dissipative mean-field equation sharing the self-consistency structure of the Kuramoto–Ott–Antonsen system. The scalar equation is not the OA equation; it shares the positive real PLS self-consistency relation, and hence the corresponding K_c, α^*, and r^* branch, but has different transient dynamics. The proof introduces a novel *pair bound*—a pointwise algebraic inequality lifted to a continuum double integral via Fubini—that gives $\dot{V} \le 0$ for the L^2 Lyapunov function without spectral theory or Fourier analysis. Global stability (no basin condition) follows from a dominated convergence bootstrap proving $r(t) \ge r_{\min} > 0$ uniformly for $K > K_c$. The entire proof chain is formalized in Lean 4 (Mathlib v4.30.0-rc1) with **0 `sorry`** and **0 project axioms**. A conditional extension to the standard complex OA equation and full Kuramoto PDE is also formalized (0 sorry, 2 cited axioms: nonlinear Landau damping and OA manifold attractivity). To the best of our knowledge, this is the first Lean formalization of a stability theorem for a Kuramoto-type model.

Main contributions.

(1) **Pair bound.** A pointwise algebraic pair inequality (Proposition 2.2) proving $\dot{V} \le 0$ for the scalar dissipative model (2). No precedent in the Kuramoto literature.

(2) **Global stability without basin condition.** A DCT bootstrap (`r_stays_positive_supercritical`, 0 sorry) proves $r(t) \ge r_{\min} > 0$ for $K > K_c$, removing the $V(0) < (r^*)^2$ assumption.

(3) **Machine-checked formalization.** The full proof chain is verified in Lean 4 with 0 sorry, 0 axioms. Coverage includes Gaussian, Student-t ($\nu > 1$), compact support, and mixtures whenever $\gamma(\omega) \lesssim 1 + |\omega|$ (ensuring $\int \gamma \, \mathrm{d}\mu < \infty$).

(4) **Conditional complex OA/PDE bridge.** Rotation cancellation in the Lyapunov derivative plus two external axioms gives $\|\eta(t)\| \to r^*$ for the standard complex OA and full Kuramoto PDE.

Contents

1 Setup and statement

1.1 The standard complex Ott–Antonsen equation

On the OA manifold, the Kuramoto–Sakaguchi PDE reduces to the *complex* OA equation:

$$\dot{z}(\omega, t) = -i\omega\, z + \frac{K}{2}\big(\bar{\eta} - \eta\, z^2\big), \qquad \eta(t) = \int_{\mathbb{R}} \overline{z(\omega, t)}\, g(\omega)\, \mathrm{d}\omega, \tag{1}$$

where $z(\omega, t) \in \mathbb{D}$ (open unit disk) and $\eta(t) \in \mathbb{C}$ is the complex order parameter. This is the standard OA reduction [5].

For symmetric g with symmetric initial data ($z(\omega, 0) = \overline{z(-\omega, 0)}$), the symmetry is preserved for all time and $\eta(t) = r(t) \in \mathbb{R}$ (`complex_oa_symmetry_preserved`, `eta_real_for_symmetric_flow`; both 0 sorry). For the Lorentzian $g(\omega) = \gamma/(\pi(\omega^2 + \gamma^2))$, residue evaluation at the pole $\omega = -i\gamma$ gives the scalar ODE $\dot{r} = (K/2 - \gamma)r - (K/2)r^3$.

1.2 A scalar dissipative model

We also study the scalar real mean-field equation

$$\dot{\alpha}(\omega, t) = -\gamma(\omega)\,\alpha + \frac{K}{2}\,r(t)\big(1-\alpha^2\big), \qquad r(t) = \int_{\mathbb{R}} \alpha(\omega, t)\,g(\omega)\,\mathrm{d}\omega, \tag{2}$$

where $\alpha(\omega, t) \in (0, 1)$ and $\gamma(\omega) > 0$ is a dissipation rate. Throughout, we write $\mathrm{d}\mu = g(\omega)\,\mathrm{d}\omega$ for the probability measure with density g; integrals $\int f\,\mathrm{d}\mu$ and $\int f\,g\,\mathrm{d}\omega$ are used interchangeably. We assume $g \geq 0$, $\int g = 1$, and g measurable.

This is a *separate dynamical system* from (1): the term $-\gamma(\omega)\alpha$ is dissipative, whereas $-i\omega z$ in (1) is rotational. The scalar model is designed to share the same positive real self-consistency equation as the symmetric partially locked state, but its transient dynamics differ from the standard complex OA equation.

Remark 1.1 (Scope clarification)**.** *Equation* (2) *is **not** derived from the complex OA equation* (1) *by setting* $z = \alpha \in \mathbb{R}$. *The subspace* $\mathrm{Im}(z) = 0$ *is not invariant under* (1) *(since* $\dot{y} = -\omega x \neq 0$ *in general). The Lyapunov argument in Sections 2–2.6 applies directly to* (2)*. The conditional extension to the standard complex OA* (1) *is given in Section 6.*

Model	Equation	Unconditional?	Lean status	Relation to Kuramoto
Scalar dissipative	Eq. (2)	Yes	0 sorry, 0 axioms	Shares PLS self-consistency
Lorentzian ODE	Bernoulli	Yes	0 sorry, 0 axioms	Classical OA for Lorentzian g
n-pole	Finite scalar	Yes	0 sorry, 0 axioms	Discrete analogue
Complex OA	Eq. (1)	No (1 axiom)	0 sorry, 1 axiom	Standard OA reduction
Full Kuramoto PDE	Kinetic	No (2 axioms)	0 sorry, 2 axioms	Original model

1.3 The partially locked state (PLS)

Assume there exists a unique positive equilibrium (α^*, r^*) satisfying the self-consistency equation

$$r^* = \int_{\mathbb{R}} \alpha^*(\omega)\,g(\omega)\,\mathrm{d}\omega \tag{3}$$

with $r^* \in (0, 1)$. (For the Lorentzian $g(\omega) = \gamma/(\pi(\omega^2 + \gamma^2))$, this holds when $K > K_c := 2\gamma$.) The equilibrium profile $\alpha^*(\omega) \in (0, 1)$ satisfies the pointwise equilibrium condition

$$\gamma(\omega)\,\alpha^*(\omega) = \frac{K}{2}\,r^*\big(1 - (\alpha^*(\omega))^2\big) \qquad \text{for } g\text{-a.e. } \omega. \tag{4}$$

1.4 Main theorem

Theorem 1.2 (Basin stability, `real_scalar_complete`)**.** *Let* $\gamma(\omega) > 0$ *with* $\int \gamma\,\mathrm{d}\mu < \infty$ *(finite first moment). Let* $(\alpha^*(\omega), r^*)$ *satisfy* (4) *and* (3)*. Let* (r, α) *solve* (2)*:*

(i) $\alpha(\omega, \cdot)$ *solves* (2) *for all* ω*, with* $r(t) = \int \alpha(\omega, t)\,\mathrm{d}\mu$*;*

(ii) $\alpha(\omega, t) \in (0, 1)$ *for all* $\omega, t \geq 0$ *(invariant region);*

(iii) $V(0) := \int (\alpha(\omega, 0) - \alpha^*(\omega))^2\,\mathrm{d}\mu < (r^*)^2$ *(initial closeness to PLS);*

(iv) for each $M > 0$, $\exists\,\delta_0 > 0$ *such that* $\alpha(\omega, 0) \geq \delta_0$ *for all* ω *with* $\gamma(\omega) \leq M$ *(initial body coherence).*

Then $r(t) \to r^*$ *as* $t \to +\infty$.

Remark 1.3. *No persistence hypothesis is assumed. The proof* derives *order parameter persistence ($r(t) \geq r_{\min} > 0$) from V antitonicity and Cauchy–Schwarz, then derives body persistence on $\{\gamma \leq M\}$ for each $M > 0$ from the ODE scalar barrier. The only conditions beyond the system definition are (iii) initial closeness and (iv) initial body coherence—both hold for any initial data in a neighborhood of the PLS.*

Theorem 1.4 (Global scalar stability, `r_stays_positive_supercritical` + `hPsi_floor_of_r_liminf`). *Let $\gamma(\omega) > 0$ with $\int \gamma \, \mathrm{d}\mu < \infty$ and $\int (1/\gamma) \, \mathrm{d}\mu < \infty$. Let (α^*, r^*) be the unique positive equilibrium satisfying (4)–(3). Let (r, α) solve (2) with $r(t) = \int \alpha(\omega, t) \, \mathrm{d}\mu$, $\alpha(\omega, t) \in (0, 1)$ for all $\omega, t \geq 0$, and initial body coherence (for each $M > 0$, $\exists \delta_0 > 0$ with $\alpha(\omega, 0) \geq \delta_0$ on $\{\gamma \leq M\}$). Assume $K > K_c := 2/\int (1/\gamma) \, \mathrm{d}\mu$.*

*Then $r(t) \to r^*$ as $t \to +\infty$, **without** assuming the basin condition $V(0) < (r^*)^2$.*

Remark 1.5. *The condition $K > K_c$ does not by itself guarantee existence of (α^*, r^*); existence is an additional hypothesis (proved for specific families such as Gaussian and Student-t via the implicit function theorem on the self-consistency equation). The scalar dissipative equation* (2) *is **not** the standard complex OA equation* (1)*; it shares the PLS self-consistency relation but has different transient dynamics.*

2 The proof chain

We present the seven-step proof of Theorem 1.2 in the order that Lean verifies it. Each step is stated as a proposition and corresponds to a named lemma in the Lean formalization.

2.1 Step 1: Leibniz rule for the Lyapunov function

Define the continuum L^2 Lyapunov function

$$V(t) := \int_{\mathbb{R}} (\alpha(\omega, t) - \alpha^*(\omega))^2 \, g(\omega) \, \mathrm{d}\omega. \tag{5}$$

Proposition 2.1 (Leibniz, `leibniz_oa_lyapunov`). *Under the hypotheses of Theorem 1.2, V is continuous on $[0, \infty)$ and for every $t > 0$:*

$$V'(t) = \int_{\mathbb{R}} 2 \, (\alpha(\omega, t) - \alpha^*(\omega)) \, \dot{\alpha}(\omega, t) \, g(\omega) \, \mathrm{d}\omega.$$

Proof. By the chain rule, $\frac{\partial}{\partial t}(\alpha(\omega, t) - \alpha^*(\omega))^2 = 2(\alpha - \alpha^*)\dot{\alpha}$. This pointwise derivative is bounded in absolute value by the integrable dominator

$$\left|2(\alpha - \alpha^*)\dot{\alpha}\right| \leq 2 \, |\dot{\alpha}| \leq 2(\gamma(\omega) + K),$$

since $|\alpha - \alpha^*| \leq 1$ and $|\dot{\alpha}| = |\gamma(\omega)\alpha - (K/2)r(1 - \alpha^2)| \leq \gamma(\omega) + K$ for $\alpha, r \in (0, 1)$. The dominator $2(\gamma(\omega) + K)$ is integrable since $\int \gamma \, g \, \mathrm{d}\omega < \infty$ by hypothesis. Mathlib's `hasDerivAt_integral_of_dominated_loc` then gives the stated formula for the derivative of $t \mapsto \int (\alpha(\omega, t) - \alpha^*)^2 \, g \, \mathrm{d}\omega$. □

2.2 Step 2: Pair bound and derived monotonicity

We derive $V'(t) \leq 0$ from the pair bound. Convergence $V(t) \to 0$ is obtained later by body-tail splitting and Barbalat (Proposition 2.6).

Proposition 2.2 (Pair bound, `pair_bound`). *For any $\alpha_j, \alpha_k, \alpha_j^*, \alpha_k^* \in (0, 1)$:*

$$\alpha_j^*(\alpha_k - \alpha_k^*)^2 \Big(\alpha_k + \frac{1}{\alpha_k^*}\Big) + \alpha_k^*(\alpha_j - \alpha_j^*)^2 \Big(\alpha_j + \frac{1}{\alpha_j^*}\Big) \geq (\alpha_j - \alpha_j^*)(\alpha_k - \alpha_k^*)(2 - \alpha_j^2 - \alpha_k^2). \tag{6}$$

Proof. Set $p_j = \alpha_j - \alpha_j^*$, $p_k = \alpha_k - \alpha_k^*$. Since $\alpha_j^*, \alpha_k^* > 0$, it suffices to show $\alpha_j^*\alpha_k^* \cdot (\mathrm{LHS} - \mathrm{RHS}) \geq 0$. Expanding the LHS terms and collecting:

$$\begin{aligned}\alpha_j^*\alpha_k^* \cdot (\mathrm{LHS} - \mathrm{RHS}) &= \underbrace{(\alpha_j^* p_k - \alpha_k^* p_j)^2}_{T_1} + \underbrace{(\alpha_j^*)^2 \alpha_k \alpha_k^*\, p_k^2}_{T_2} \\ &\quad + \underbrace{(\alpha_k^*)^2 \alpha_j \alpha_j^*\, p_j^2}_{T_3} + \underbrace{\alpha_j^*\alpha_k^*(\alpha_j^2 + \alpha_k^2)\, p_j p_k}_{T_4}. \end{aligned} \tag{7}$$

Case $p_j p_k \leq 0$. Then LHS ≥ 0 (each summand is (pos) $\cdot\, p^2 \cdot$ (pos)) and RHS $= p_j p_k(2 - \alpha_j^2 - \alpha_k^2) \leq 0$ (since $2 - \alpha_j^2 - \alpha_k^2 > 0$ for $\alpha_j, \alpha_k \in (0,1)$). Hence LHS $\geq 0 \geq$ RHS.

Case $p_j p_k > 0$. Terms $T_1, T_2, T_3 \geq 0$ and $T_4 > 0$. We show $T_2 + T_3 + T_4 \geq 0$, which suffices since $T_1 \geq 0$. Since $p_j p_k > 0$, both have the same sign. Write $\alpha_j = \alpha_j^* + p_j$ and $\alpha_k = \alpha_k^* + p_k$:

$$\begin{aligned} T_2 + T_3 + T_4 &= (\alpha_j^*)^2(\alpha_k^* + p_k)\alpha_k^*\, p_k^2 + (\alpha_k^*)^2(\alpha_j^* + p_j)\alpha_j^*\, p_j^2 \\ &\quad + \alpha_j^*\alpha_k^*\big((\alpha_j^* + p_j)^2 + (\alpha_k^* + p_k)^2\big)p_j p_k. \end{aligned}$$

Since $\alpha_j, \alpha_k \in (0,1)$ we have $\alpha_j = \alpha_j^* + p_j > 0$ and $\alpha_k = \alpha_k^* + p_k > 0$, so all factors involving α_j, α_k are positive. With $p_j p_k > 0$, every term in $T_2 + T_3 + T_4$ is non-negative (using $\alpha_j^* + p_j = \alpha_j > 0$ for T_2's factor $\alpha_k = \alpha_k^* + p_k > 0$, and similarly for T_3). Hence $T_2 + T_3 + T_4 > 0$ and the decomposition (7) is positive.

The full expansion is verified in Lean as `pair_bound_clear_denom` (0 sorry). ☐

Proposition 2.3 (Continuum pair bound, `continuum_pair_nonneg`)**.** *The double integral*

$$\begin{aligned}\int_{\mathbb{R}}\int_{\mathbb{R}} \Big[&\alpha^*(\omega_1)(\alpha(\omega_2) - \alpha^*(\omega_2))^2\Big(\alpha(\omega_2) + \frac{1}{\alpha^*(\omega_2)}\Big) + \alpha^*(\omega_2)(\alpha(\omega_1) - \alpha^*(\omega_1))^2\Big(\alpha(\omega_1) + \frac{1}{\alpha^*(\omega_1)}\Big) \\ &-(\alpha(\omega_1) - \alpha^*(\omega_1))(\alpha(\omega_2) - \alpha^*(\omega_2))(2 - \alpha(\omega_1)^2 - \alpha(\omega_2)^2)\Big]\, g(\omega_1)\, g(\omega_2)\, \mathrm{d}\omega_1\, \mathrm{d}\omega_2 \geq 0.\end{aligned}$$

Proof. Mathlib's `integral_nonneg` applied twice (once in ω_1, once in ω_2) reduces this to the pointwise inequality of Proposition 2.2. ☐

Proposition 2.4 (Derived monotonicity from the pair bound, `hV_rate`)**.** *Under the hypotheses of Theorem 1.2, $V'(t) \leq 0$ for all $t > 0$. Convergence $V \to 0$ follows from body persistence combined with Barbalat's lemma via the body-tail split: on $\{\gamma \leq M\}$, body persistence gives a coercive lower bound on $-V'$; on $\{\gamma > M\}$, the tail contribution vanishes as $M \to \infty$ since $\int \gamma\, g < \infty$. This is* proved *from the pair bound and persistence; it is not assumed.*

Proof sketch. Substituting the OA equation (2) into the Leibniz formula:

$$V'(t) = \int_{\mathbb{R}} 2(\alpha - \alpha^*)\Big[-\gamma\alpha + \frac{K}{2} r(1 - \alpha^2)\Big] g\, \mathrm{d}\omega.$$

Using the equilibrium condition (4) to write $\gamma\alpha^* = (K/2)r^*(1 - (\alpha^*)^2)$, one computes:

$$2(\alpha - \alpha^*)\dot{\alpha} = K(\alpha - \alpha^*)\Big[r(1 - \alpha^2) - r^*(1 - (\alpha^*)^2) - \frac{2\gamma}{K}(\alpha - \alpha^*)\Big].$$

Expanding and grouping, the integrand splits into:

- a term proportional to $(r - r^*) \cdot D$ where $D = \int(\alpha - \alpha^*)g\, \mathrm{d}\omega = r - r^*$, giving $K(r - r^*)^2 \leq KV$ (by Cauchy-Schwarz);
- a term $-Kr^* \cdot Q$ where $Q = \int(\alpha - \alpha^*)^2(\alpha + 1/\alpha^*)g\, \mathrm{d}\omega$;
- a term $K \cdot DS$ where $S = \int(\alpha - \alpha^*)(1 - \alpha^2)g\, \mathrm{d}\omega$.

The continuum pair bound (Proposition 2.3) gives $DS \le r^*Q$, so $K \cdot DS - Kr^*Q \le 0$, and the pair bound already yields $V'(t) \le 0$. To obtain convergence $V \to 0$, one uses the body-tail split: on the body $\{\gamma \le M\}$, body persistence gives $\alpha(\omega,t) \ge \delta(M) > 0$, which provides a coercive lower bound $Q_{\text{body}} \ge \delta(M) \cdot \delta^* \cdot V_{\text{body}}$; the tail $\{\gamma > M\}$ contributes at most $\int_{\gamma>M} \gamma\, g\, \mathrm{d}\omega \to 0$ as $M \to \infty$. Barbalat's lemma then gives $V \to 0$. The full computation is verified in `ContinuumUniformRate.lean`. □

2.3 Step 3: V is antitone

Proposition 2.5 (V antitone, `lyapunov_antitone`)**.** *$V(t)$ is non-increasing on $[0,\infty)$.*

Proof. By Proposition 2.4, $V'(t) \le 0$ for all $t > 0$. Since V is continuous (Proposition 2.1), Mathlib's `antitoneOn_of_deriv_nonpos` applies directly on $[0,\infty)$. □

2.4 Step 4: Body-tail split and Barbalat

Proposition 2.6 (Body-tail convergence, `supercritical_coercive_drops`)**.** *For any $\varepsilon > 0$, there exists $T > 0$ such that $V(t) < \varepsilon$ for all $t > T$.*

Proof. **Step A (Tail control).** Decompose

$$V = V_{\text{body},M} + V_{\text{tail},M}, \quad V_{\text{body},M} := \int_{\gamma\le M} (\alpha-\alpha^*)^2\, g, \quad V_{\text{tail},M} := \int_{\gamma>M} (\alpha-\alpha^*)^2\, g.$$

Since $|\alpha - \alpha^*| \le 1$ and $(\alpha-\alpha^*)^2 \le 1$:

$$V_{\text{tail},M} \le \int_{\gamma>M} g\, \mathrm{d}\omega.$$

Since $\int \gamma\, g < \infty$, Markov's inequality gives $\int_{\gamma>M} g \le \frac{1}{M}\int \gamma\, g \to 0$ as $M \to \infty$. Thus for every $\varepsilon > 0$, there exists M_0 such that $V_{\text{tail},M_0}(t) < \varepsilon/2$ *for all t.*

Step B (Body convergence). Fix $M = M_0$. On $\{\gamma \le M\}$, body persistence (derived in Step 3 of the proof of Theorem 1.2) gives $\alpha(\omega,t) \ge \delta(M) > 0$. The body-restricted pair coercivity then gives a positive lower bound:

$$-V'(t) \ge K \cdot \delta(M) \cdot \delta^*(M) \cdot \mu(\gamma \le M) \cdot V_{\text{body},M}(t),$$

where $\delta^*(M) = \min_{\gamma\le M} \alpha^*(\omega) > 0$. Since V is antitone and $V' \le -c \cdot V_{\text{body},M}$ with $c > 0$: the time-integral $\int_0^\infty V_{\text{body},M}(t)\,\mathrm{d}t$ converges (bounded by $V(0)/c$). The derivative of $V_{\text{body},M}$ satisfies

$$\left|\frac{d}{dt} V_{\text{body},M}(t)\right| = \left|\int_{\gamma\le M} 2(\alpha-\alpha^*)\dot{\alpha}\, g\right| \le 2(M+K)\,\mu(\gamma \le M),$$

since $|\alpha-\alpha^*| \le 1$ and $|\dot\alpha| \le M + K$ on $\{\gamma \le M\}$. A bounded derivative gives Lipschitz, hence uniform continuity of $V_{\text{body},M}$. Since $V_{\text{body},M} \ge 0$ and $\int_0^\infty V_{\text{body},M}(t)\,\mathrm{d}t < \infty$, Barbalat's lemma gives $V_{\text{body},M}(t) \to 0$.

Step C (Combine). For t large enough: $V_{\text{body},M_0}(t) < \varepsilon/2$ (Step B) and $V_{\text{tail},M_0}(t) < \varepsilon/2$ (Step A). Therefore $V(t) < \varepsilon$.

This avoids any appeal to uniform continuity of V'. The Lean formalization uses `coercive_drop_from_persi` in `ContinuumBarbalat.lean`, which takes: non-negativity, antitonicity, body persistence, and tail vanishing. □

Remark 2.7. *In the n-pole case (Section 5), the stronger exponential rate $V(t+1) \le e^{-\mu}V(t)$ with $\mu = Kc_{\min}\delta\delta^*$ holds via Gronwall comparison (`comparison_decay_interval` in `GronwallBridge.lean`).*

2.5 Step 5: $V \to 0$

Proposition 2.8 ($V \to 0$, `coercive_drop_from_persistence`)**.** *$V(t) \to 0$ as $t \to +\infty$.*

Proof. By Proposition 2.6, $V(t) \to 0$ via the body-tail split and Barbalat's lemma. Since V is antitone (Proposition 2.5) and bounded below by 0, the limit exists, and the body-tail argument forces it to be zero. The formalization uses `coercive_drop_from_persistence` in `ContinuumBarbalat.lean`, which takes: non-negativity, antitonicity, body persistence, and tail vanishing. □

2.6 Step 6: $r \to r^*$ via Cauchy-Schwarz

Proposition 2.9 ($r \to r^*$, closing step)**.** *$V(t) \to 0$ implies $r(t) \to r^*$.*

Proof. By the self-consistency relation and the equilibrium formula:

$$r(t) - r^* = \int_{\mathbb{R}} (\alpha(\omega, t) - \alpha^*(\omega))\, g(\omega)\, \mathrm{d}\omega.$$

By the Cauchy-Schwarz inequality (or Jensen applied to the square function):

$$(r(t) - r^*)^2 = \left(\int_{\mathbb{R}} (\alpha - \alpha^*)\, g\, \mathrm{d}\omega \right)^2 \leq \int_{\mathbb{R}} (\alpha - \alpha^*)^2\, g\, \mathrm{d}\omega = V(t).$$

This is formalized as `sq_integral_le_integral_sq` in the Lean code. Since $V(t) \to 0$, we have $(r(t) - r^*)^2 \to 0$, hence $r(t) \to r^*$ in the metric sense. □

Proof of Theorem 1.2. Steps 1 through 6 above, composed in order, constitute the complete proof. In Lean:

(1) `leibniz_oa_lyapunov` gives continuity and differentiability of V.

(2) The pair bound (`continuum_pair_nonneg`) and Leibniz together give $V' \leq 0$.

(3) `lyapunov_antitone` gives that V is non-increasing.

(4) Cauchy–Schwarz gives $r(t) \geq r^* - \sqrt{V(0)} > 0$ (order parameter persistence).

(5) ODE scalar barrier gives body persistence on $\{\gamma \leq M\}$ for each $M > 0$.

(6) `coercive_drop_from_persistence` gives $V \to 0$ via body-tail split + Barbalat.

(7) `sq_integral_le_integral_sq` gives $(r - r^*)^2 \leq V$, so $r \to r^*$.

The end-to-end theorem is `real_scalar_complete` in `RealScalarComplete.lean`, which wires the chain and derives body persistence internally from the basin condition. □

2.7 Step 7: Pair rigidity (characterization of $V' = 0$)

Although not needed for the convergence proof itself, the pair bound also characterizes when $V' = 0$.

Proposition 2.10 (Pair rigidity, `pair_eq_zero_iff`)**.** *The pair integrand* (6) *equals zero if and only if $\alpha_j = \alpha_j^*$ and $\alpha_k = \alpha_k^*$.*

Proof. The algebraic decomposition plus sign case split in the proof of Proposition 2.2 shows:

$$\alpha_j^*\alpha_k^* \cdot (\text{LHS} - \text{RHS}) = (\alpha_j^* p_k - \alpha_k^* p_j)^2 + (\alpha_j^*)^2\alpha_k\alpha_k^* p_k^2 + (\alpha_k^*)^2\alpha_j\alpha_j^* p_j^2 + \alpha_j^*\alpha_k^*(\alpha_j^2 + \alpha_k^2)p_j p_k.$$

When $p_j p_k > 0$: the second and third terms are strictly positive, so the sum is positive and the integrand cannot be zero unless $p_j = p_k = 0$. When $p_j p_k \le 0$: each of the three original terms of (6) is non-negative (since $\alpha_j + 1/\alpha_j^* > 0$ and $2 - \alpha_j^2 - \alpha_k^2 > 0$), so all must individually vanish, giving $p_j = p_k = 0$. This is formalized as `pair_eq_zero_iff` in `ContinuumLyapunov.lean`. □

Remark 2.11 (LaSalle structure). *Proposition 2.10 shows that $\{V' = 0\} = \{\alpha = \alpha^*\}$. This is the LaSalle invariance principle characterization and is used in the alternative Path B proof (`ContinuumGlobalStability.lean`) where scalar asymptotic autonomy of each oscillator gives pointwise convergence and dominated convergence gives $V \to 0$.*

2.8 Step 8: Removing the basin condition (DCT bootstrap)

This step is not used in Theorem 1.2 (which assumes the basin condition) but is the key additional ingredient for Theorem 1.4.

Proposition 2.12 (Uniform positivity, `r_stays_positive_supercritical`). *Under hypotheses (i), (ii), (iv) of Theorem 1.2, if $K > K_c := 2/\int(1/\gamma)\,\mathrm{d}\mu$ and $\int(1/\gamma)\,\mathrm{d}\mu < \infty$, then there exists $r_{\min} > 0$ such that $r(t) \ge r_{\min}$ for all $t \ge 0$.*

Proof. **Step A (DCT extraction).** Define the approximating sequence

$$G_n(\omega) := \min\big((n{+}1)\cdot\alpha(\omega,0),\ K/(2\gamma(\omega) + K/(n{+}1))\big).$$

Each G_n is dominated by $K/(2\gamma(\omega))$, which is integrable (since $\int(1/\gamma)\,\mathrm{d}\mu < \infty$). Since $\alpha(\omega,0) > 0$ and $\gamma(\omega) > 0$, pointwise $G_n(\omega) \to K/(2\gamma(\omega))$ as $n \to \infty$. The dominated convergence theorem gives $\int G_n\,\mathrm{d}\mu \to \int K/(2\gamma)\,\mathrm{d}\mu = (K/2)\int(1/\gamma)\,\mathrm{d}\mu > 1$ (by $K > K_c$).

Extract N with $\int G_N\,\mathrm{d}\mu > 1$ and $1/(N{+}1) < r(0)$. Set $\varepsilon_0 := 1/(N{+}1)$. Then $\varepsilon_0 < r(0)$, $\varepsilon_0 \le 1$. Dividing G_N by $(N{+}1)$:

$$\frac{G_N(\omega)}{N{+}1} = \min\big(\alpha(\omega,0),\ K\varepsilon_0/(2\gamma(\omega) + K\varepsilon_0)\big),$$

so $\int G_N\,\mathrm{d}\mu > 1$ gives $\varepsilon_0 = 1/(N{+}1) < (1/(N{+}1))\int G_N\,\mathrm{d}\mu = \int \min(\alpha(\omega,0), K\varepsilon_0/(2\gamma + K\varepsilon_0))\,\mathrm{d}\mu$.

Step B (Self-improvement). If $r(s) \ge \varepsilon_0$ for all $s \in [0,T]$, then for each ω: either $\alpha(\omega,0) \ge \beta^*(\gamma(\omega),K,\varepsilon_0)$ (the body equilibrium), in which case the ODE barrier gives $\alpha(\omega,T) \ge \beta^* \ge K\varepsilon_0/(2\gamma(\omega) + K\varepsilon_0)$; or $\alpha(\omega,0) < \beta^*$, in which case monotonicity below equilibrium gives $\alpha(\omega,T) \ge \alpha(\omega,0)$. In either case, $\alpha(\omega,T) \ge \min(\alpha(\omega,0), K\varepsilon_0/(2\gamma(\omega) + K\varepsilon_0))$. Integrating: $r(T) = \int \alpha(\omega,T)\,\mathrm{d}\mu > \varepsilon_0$.

Step C (Contradiction closure). Suppose $r(t_0) < \varepsilon_0$ for some $t_0 > 0$. The set $\{t \ge 0 : r(t) \le \varepsilon_0\}$ is closed (continuity of r) and bounded below; let $T = \inf\{t \ge 0 : r(t) \le \varepsilon_0\}$. Then $T > 0$ (since $r(0) > \varepsilon_0$), $r(T) = \varepsilon_0$, and $r(s) \ge \varepsilon_0$ for all $s \in [0,T]$. Step B gives $r(T) > \varepsilon_0$, contradicting $r(T) = \varepsilon_0$. □

Proof of Theorem 1.4. Proposition 2.12 gives $r(t) \ge r_{\min} > 0$ for all $t \ge 0$. This uniform positivity provides the persistence hypotheses needed in the body-tail/Barbalat argument (Proposition 2.6): on $\{\gamma \le M\}$, the ODE barrier gives $\alpha(\omega,t) \ge \delta(M) > 0$ for all t, using $r \ge r_{\min}$ in place of the Cauchy–Schwarz bound from the basin condition. The body-tail split then gives $V(t) \to 0$ (Proposition 2.8). Finally, Proposition 2.9 gives $r(t) \to r^*$.

In the Lean formalization, this is assembled as `hPsi_floor_of_r_liminf`, which reuses the body persistence + Barbalat infrastructure of `kuramoto_V_zero_of_r_floor` (a shared subroutine, not a circular invocation of Theorem 1.2). □

3 The Lorentzian case: fully end-to-end

For the Lorentzian distribution $g(\omega) = \gamma/(\pi(\omega^2 + \gamma^2))$, the OA equation reduces to the scalar Bernoulli ODE

$$\dot{r} = \left(\frac{K}{2} - \gamma\right) r - \frac{K}{2} r^3. \tag{8}$$

The equilibrium is $r^* = \sqrt{1 - 2\gamma/K}$ for $K > 2\gamma$.

Theorem 3.1 (Lorentzian convergence, `lorentzian_ode_global_stability_complete`). *For $K > 2\gamma > 0$ and any $r_0 \in (0, 1)$, the unique solution of (8) satisfies $r(t) \to r^*$.*

The proof chain in Lean is:

$(K, \gamma, r_0) \xrightarrow{\text{Bernoulli}}$ `LorentzianContinuousSolution` $\to$ `LorentzianSolution` $\to$ `KuramotoData` $\to$ $r(t)$ -

3.1 Proof: Bernoulli formula

Proposition 3.2 (Existence via Bernoulli, `LorentzianExistence`). *The ODE (8) has an explicit solution $r(t) = (w(t))^{-1/2}$ where $w(t) = w_0 e^{-(K-2\gamma)t} + (K/(K - 2\gamma))(1 - e^{-(K-2\gamma)t})$ and $w_0 = r_0^{-2}$. This function satisfies `HasDerivAt` at every $t \geq 0$, verified by differentiating the explicit formula.*

3.2 Proof: Monotonicity

Proposition 3.3 (Monotonicity, `LorentzianFromODE`). *If $r_0 < r^*$ then r is non-decreasing. If $r_0 > r^*$ then r is non-increasing. In both cases $r \in (0, 1)$ for all t.*

Proof. From (8), $\dot{r} = (K/2)r(1 - 2\gamma/K - r^2) = (K/2)r((r^*)^2 - r^2)$. So $\dot{r} > 0$ when $r < r^*$ and $\dot{r} < 0$ when $r > r^*$. Invariance $r \in (0, 1)$ follows from the barrier structure at $r = 0$ and $r = 1$. □

3.3 Proof: Convergence

Proof of Theorem 3.1. r is monotone and bounded, so $r(t) \to L$ for some $L \in [0, 1]$. Taking $t \to \infty$ in (8) (using continuity of the right side and $\dot{r} \to 0$ from the monotone convergence theorem): $0 = (K/2 - \gamma)L - (K/2)L^3$. The nonzero solutions are $L = \pm r^*$; since $r > 0$ always, $L = r^*$. □

The complete proof is assembled in `lorentzian_ode_global_stability_complete` with zero additional axioms beyond Lean's standard foundations, verified in `LorentzianInstance.lean`.

4 Lean 4 verification

4.1 Overall status

The formalization spans **255 Lean 4 files** (commit `58b6194`, Mathlib v4.30.0-rc1). The declaration `real_scalar_complete` depends on no project axioms and no `sorry`; the imported development contains unrelated `sorry`s in theorems outside this dependency chain. The formalization uses only Lean 4's standard foundations.

There are three tiers of completeness:

Tier 1 — Lorentzian (fully end-to-end). The theorem `lorentzian_ode_global_stability_complete` takes only the physical parameters (K, γ, r_0) and constructs the complete proof: solution existence via the explicit Bernoulli formula, all solution properties, and convergence $r \to r^*$. Zero axioms, zero assumed fields.

Tier 2 — Finite first moment (fully self-contained). The theorem `kuramoto_stability` in `KuramotoFinal.lean` derives everything internally:

1. Leibniz rule $\to V$ differentiable (finite first moment provides the integrable dominator $2(\gamma(\omega) + K)$)
2. Pair bound $\to \dot{V} \leq 0$
3. V antitone $\to V(t) \leq V(0) < (r^*)^2$
4. Cauchy–Schwarz $\to r(t) \geq r^* - \sqrt{V(0)} > 0$ (order parameter persistence)
5. ODE scalar barrier $\to$ body persistence on $\{\gamma \leq M\}$ for each $M > 0$
6. Body-tail split + Barbalat $\to V \to 0 \to r(t) \to r^*$

No persistence hypothesis is assumed externally. No uniform-in-ω lower bound is assumed or derived. The only initial conditions are $V(0) < (r^*)^2$ and body coherence at $t = 0$. **0 sorry, 0 axioms.**

Tier 3 — N-pole (any finite n). The theorem `kuramoto_solved` provides an alternative proof path for discrete approximations with explicit exponential rate $\dot{V} \leq -\mu V$ where $\mu = K c_{\min} \delta \delta^*$.

4.2 The proof chain in Lean

The following table shows how each proposition in Section 2 corresponds to a Lean theorem:

Proposition	**Lean name**	**File**
Leibniz rule (Prop. 2.1)	`leibniz_oa_lyapunov`	`GeneralGMainTheorem.lean`
Pair bound (Prop. 2.2)	`pair_bound`	`L2Lyapunov.lean`
Continuum pair (Prop. 2.3)	`continuum_pair_nonneg`	`ContinuumLyapunov.lean`
Monotonicity bound (Prop. 2.4)	`hV_rate`	`ContinuumUniformRate.lean`
V antitone (Prop. 2.5)	`lyapunov_antitone`	`GeneralGMainTheorem.lean`
Exp. drop (Prop. 2.6)	`supercritical_coercive_drops`	`GeneralGMainTheorem.lean`
$V \to 0$ (Prop. 2.8)	`coercive_drop_from_persistence`	`ContinuumBarbalat.lean`
$r \to r^*$ (Prop. 2.9)	`sq_integral_le_integral_sq`	`GeneralGMainTheorem.lean`
Pair rigidity (Prop. 2.10)	`pair_eq_zero_iff`	`ContinuumLyapunov.lean`

Key Mathlib lemmas used in the proof chain:

- `hasDerivAt_integral_of_dominated_loc_of_deriv_le` — Leibniz differentiation under the integral sign;
- `antitoneOn_of_deriv_nonpos` — monotone convergence from a derivative bound;
- `integral_nonneg` — applied pointwise to transfer the pair bound to integrals;
- `MeasureTheory.continuous_of_dominated` — dominated convergence for continuity of V.

4.3 Nine independent proof paths

The formalization contains nine complementary, sorry-free proof paths of basin stability or closely related statements. Each addresses a different mathematical angle:

File	Proof strategy
`GeneralGMainTheorem`	Leibniz + pair bound + coercive Barbalat (Sections 2.1–2.6; this paper)
`MainTheorem`	Gap exclusion + Lipschitz trapping (discrete, self-consistency)
`ContinuousStability`	IVT trapping (continuous-time)
`NPoleGlobalStability`	L^2 Barbalat + persistence drops (n-pole systems)
`GronwallBridge` + `NPoleInstance`	L^2 Gronwall + exponential rate (continuous n-pole)
`ContinuumLyapunov`	$\dot{V} \leq 0$ directly (pair bound $\to$ integrals)
`AntitoneConvergence`	$V \to L \geq 0$ (bounded antitone sequences)
`ContinuumGlobalStability`	Scalar asymptotic autonomy (Path B: each $\alpha(\omega, \cdot) \to \alpha^*(\omega)$)
`InstabilityExclusion`	V antitone + instability escape $\Rightarrow V \to 0$ (no persistence needed)

4.4 Additional formalized results

Beyond global stability, the formalization establishes:

- **Complete trifurcation**: $K < K_c \Rightarrow r(t) \to 0$; $K = K_c \Rightarrow r(t) \to 0$ at rate $O(1/\sqrt{t})$; $K > K_c \Rightarrow r(t) \to r^*$.
- **Bifurcation monotonicity**: $r^*(K)$ is increasing in K for $K > K_c$.
- **Instability of incoherence**: proved via Chetaev's theorem for $K > K_c$.
- **Lorentzian trifurcation**: $K < 2\gamma \Rightarrow r \to 0$, $K = 2\gamma \Rightarrow r \to 0$, $K > 2\gamma \Rightarrow r \to r^*$.

To verify: `cd kuramoto-lean && lake build KuramotoLean.RealScalarComplete` (commit `58b6194`).

5 The L^2 Lyapunov approach for n-pole systems

This section describes the cleanest of the nine proof paths for finite approximations: an explicit Lyapunov function for n-pole systems that decreases monotonically and decays exponentially. The argument is a finite-dimensional specialization of the continuum proof in Section 2.

5.1 Setup

An *n-pole system* is a finite approximation to the continuum Kuramoto model in which the frequency distribution g is replaced by a discrete measure $\sum_{k=1}^{n} c_k \, \delta_{\omega_k}$. The state is $(\alpha_k)_{k=1}^{n} \in (0, 1)^n$ and the order parameter is $r = \sum_k c_k \alpha_k$.

Let α_k^* denote the PLS values. Define the L^2 Lyapunov function:

$$V(t) = \sum_{k=1}^{n} c_k \, |\alpha_k(t) - \alpha_k^*|^2. \tag{9}$$

5.2 Main decay estimate

Theorem 5.1 (L^2 Lyapunov decay, `l2_exponential_rate`). *For any n-pole Kuramoto system with $K > K_c$, there exist constants $c_{\min} > 0$, $\delta > 0$, $\delta^* > 0$ depending only on K, g, and the pole weights such that*

$$\frac{\mathrm{d}V}{\mathrm{d}t} \le -K\, c_{\min}\, \delta\, \delta^*\, V.$$

In particular, $V(t) \le V(0)\, e^{-K c_{\min} \delta \delta^* t} \to 0$ exponentially.

Proof outline. Differentiate (9) using the Kuramoto ODEs. Each pair (α_k, α_k^*) contributes a term involving $\mathrm{Re}\big(\overline{(\alpha_k - \alpha_k^*)}(\dot\alpha_k - \dot\alpha_k^*)\big)$. Writing $\dot\alpha_k - \dot\alpha_k^*$ in terms of $r - r^*$ and using the finite sum version of the pair bound (Proposition 2.2), together with the uniform lower bound $c_{\min} = \min_k c_k$:

$$\frac{\mathrm{d}V}{\mathrm{d}t} \le -K\, c_{\min}\, \delta\, \delta^*\, V.$$

This is formalized in `UniformRate.lean` and `L2Lyapunov.lean` (0 sorry). □

Remark 5.2 (Uniformity in n). *The bound $dV/dt \le -K\, c_{\min}\, \delta\, \delta^*\, V$ is* uniform *in n: the constant does not depend on the number of poles, only on the weight floor $c_{\min}$ and the gap parameters δ, δ^*. In the continuum limit (Proposition 2.4), the explicit exponential rate is replaced by the body-tail split and Barbalat's lemma, since $c_{\min} \to 0$ as the discrete measure approximates a continuous density.*

5.3 Convergence from the Lyapunov bound

Global stability follows immediately: $V(t) \to 0$ implies $\|\alpha(t) - \alpha^*\|^2_{L^2(g)} \to 0$, which in particular forces $|r(t) - r^*| \to 0$ by Cauchy-Schwarz (Proposition 2.9). The chain is:

$$V(0) < \infty \xrightarrow{\text{Thm. 5.1}} V(t) \to 0 \xrightarrow{\text{Cauchy–Schwarz}} |r(t) - r^*| \to 0.$$

This proof path (`NPoleGlobalStability.lean`, `GronwallBridge.lean`, `NPoleInstance.lean`) requires no Barbalat lemma beyond the monotone convergence theorem, and no analyticity of g—only the existence of the PLS and the gap bound.

Remark 5.3 (Local vs. global stability). *For n-pole systems (any finite n), the theorem `kuramoto_solved` gives* global *stability: any $r(0) > 0$ converges. Persistence is automatic in finite dimensions. For the continuum scalar dissipative model* (2)*, global stability is now also proved: `r_stays_positive_supercritical` in `ContinuumInstability.lean` (0 sorry) shows that for $K > K_c$, the order parameter stays uniformly positive ($r(t) \ge r_{\min} > 0$) via a DCT bootstrap argument. The proof constructs $\varepsilon_0 > 0$ such that the self-consistency integral $\int \min(\alpha(\omega, 0),\, K\varepsilon_0/(2\gamma(\omega) + K\varepsilon_0))\, \mathrm{d}\mu > \varepsilon_0$, then uses body persistence to show r can never first cross below ε_0 (IVT + closed crossing set contradiction). This removes the basin condition $V(0) < (r^*)^2$ for the scalar dissipative model.*

6 Complex OA: rotation cancellation and conditional stability

The pair bound argument of Sections 2–2.6 applies directly to the scalar model (2). We now examine what structural elements carry over to the standard complex OA equation (1). The rotation term admits an exact cancellation in the V derivative, but the remaining nonlinear coupling terms do not reduce to the scalar pair-bound structure. The complex OA convergence is therefore *conditional* on external nonlinear Landau damping estimates.

6.1 Dietert energy identity

Proposition 6.1 (`psi_deriv_eq_K_eta_sq`, 0 sorry)**.** *Define the Dietert energy* $\Psi(t) := -\int \log(1-|z(\omega,t)|^2)\, g\, \mathrm{d}\omega$*. Then*

$$\frac{\mathrm{d}\Psi}{\mathrm{d}t} = K|\eta(t)|^2 \geq 0.$$

Proof. By chain rule, $\frac{\mathrm{d}}{\mathrm{d}t}[-\log(1-|z|^2)] = \frac{\mathrm{d}|z|^2/\mathrm{d}t}{1-|z|^2}$. From the complex OA, $\mathrm{d}|z|^2/\mathrm{d}t = K\ \mathrm{Re}(\eta z)(1-|z|^2)$ (`complexOa_normSq_deriv`, 0 sorry). Hence the pointwise integrand is $K\ \mathrm{Re}(\eta z)$. Integrating: $\mathrm{d}\Psi/\mathrm{d}t = K\int \mathrm{Re}(\eta z)\, g = K\ \mathrm{Re}(\eta\bar{\eta}) = K|\eta|^2$, using $\bar{\eta} = \int z\, g$ for real-valued g (`complex_eta_integral_identity`, 0 sorry). ☐

6.2 Rotation cancellation

The L^2 Lyapunov function $V(t) = \int |z - z^*|^2\, g\, \mathrm{d}\omega$ has derivative

$$V'(t) = 2\int \mathrm{Re}\big(\overline{(z-z^*)}\cdot \dot{z}\big)\, g\, \mathrm{d}\omega.$$

Subtracting the equilibrium equation $0 = -i\omega z^* + (K/2)(\bar{\eta}^* - \eta^* z^{*2})$:

$$\dot{z} - 0 = -i\omega(z-z^*) + \tfrac{K}{2}\big[(\eta-\eta^*)(1-z^2) + \eta^*(z^{*2}-z^2)\big].$$

Proposition 6.2 (Rotation cancellation, `rotation_zero_in_error`, 0 sorry)**.** $\mathrm{Re}\big(\overline{(z-z^*)}\cdot(-i\omega(z-z^*))\big) = \mathrm{Re}(-i\omega|z-z^*|^2) = 0.$

Proof. $|z-z^*|^2 \in \mathbb{R}$, so $-i\omega|z-z^*|^2$ is purely imaginary. ☐

The rotation cancellation removes the rotation contribution from V', but the remaining nonlinear coupling terms differ structurally from the scalar case: the coupling involves $\eta \in \mathbb{C}$ (not $r \in \mathbb{R}$), and the quadratic terms $\eta^*(z^{*2}-z^2)$ do not factor as cleanly as in the scalar pair bound. The scalar pair-bound technique does *not* extend to the complex OA. Convergence $V(t) \to 0$ is instead obtained from nonlinear Landau damping [9]: for Sobolev-regular perturbations of the PLS, phase mixing of drifting oscillators gives algebraic decay $|V(t)| = O(t^{-s})$ (stated as axiom `landau_damping_kuramoto`). This is a *conditional* result relying on external analytic input, not a self-contained machine-checked proof.

6.3 Symmetry preservation

Proposition 6.3 (`complex_oa_symmetry_preserved`, 0 sorry)**.** *For symmetric g with symmetric initial data, $z(\omega,t) = \overline{z(-\omega,t)}$ for all t. Consequently, $\eta(t) \in \mathbb{R}$.*

Proof. Define $w(\omega,t) := \overline{z(-\omega,t)}$. Then w satisfies the same *coupled* OA system: conjugating the ODE and using $g(-\omega) = g(\omega)$ gives $\dot{w}(\omega,t) = -i\omega w + (K/2)(\bar{\eta}_w - \eta_w w^2)$ where $\eta_w = \int \bar{w}\, g = \int z(-\cdot)\, g = \int z\, g$ (by measure-preserving negation) $= \overline{\eta_z}$ (by `integral_conj`). The algebraic identity `complexOaRHS_conj_neg` gives $\overline{f(-\omega,\eta,z)} = f(\omega,\bar{\eta},\bar{z})$. Since $w(\omega,0) = z(\omega,0)$ and both satisfy the same coupled system, *coupled* ODE uniqueness gives $w = z$, establishing symmetry *and* $\eta \in \mathbb{R}$ simultaneously without circularity (0 sorry). ☐

6.4 Full Kuramoto PDE stability

Theorem 6.4 (Conditional bridge lemma, `full_kuramoto_pde_stability_via_axiom`, 0 sorry + 1 axiom)**.** *For the full Kuramoto–Sakaguchi PDE with analytic symmetric g, $K > K_c$: if the OA order parameter converges ($\|\eta_{\mathrm{OA}}(t)\| \to r^*$) and the full PDE approximates the OA trajectory ($\|\eta_{\mathrm{full}} - \eta_{\mathrm{OA}}\| \to 0$), then $\|\eta_{\mathrm{full}}(t)\| \to r^*$. This theorem is conditional: both hypotheses are cited axioms, not machine-checked.*

Proof. Triangle inequality: $\big|\|\eta_{\text{full}}\| - \|\eta_{\text{OA}}\|\big| \le \|\eta_{\text{full}} - \eta_{\text{OA}}\| \to 0$. Squeeze gives convergence (`squeeze_zero_norm + abs_norm_sub_norm_le`, 0 sorry). □

Remark 6.5 (Hypotheses of Theorem 6.4)**.** *The hypothesis $\|\eta_{\text{full}} - \eta_{\text{OA}}\| \to 0$ follows from OA manifold exponential attractivity [2] (axiom). The hypothesis $\|\eta_{\text{OA}}\| \to r^*$ follows from nonlinear Landau damping [9] (axiom) giving $V \to 0$, then Cauchy–Schwarz (machine-checked). The full chain is `kuramoto_stability_universal` (0 sorry, 2 axioms).*

7 Discussion

7.1 Formalization tiers

The results are organized into four certainty classes (with Theorems A and A′ forming a single scalar tier at two levels of generality):

Theorem A: Real scalar complete (`real_scalar_complete`). Fully formalized, 0 sorry, 0 axioms. Basin stability for the scalar dissipative model (2) with finite first moment $\int \gamma\, g < \infty$. When $\gamma(\omega) \lesssim 1 + |\omega|$: Gaussian, Student-$t$ ($\nu > 1$), compact support, mixtures.

Theorem A′: Real scalar global (`r_stays_positive_supercritical` + `hPsi_floor_of_r_liminf` + `real_scalar_complete`). Fully formalized, 0 sorry, 0 axioms. **Global** stability for the scalar dissipative model: for $K > K_c := 2/\int(1/\gamma)\,\mathrm{d}\mu$ and $\alpha(\omega, 0) \in (0, 1)$, the order parameter converges $r(t) \to r^*$ *without* assuming $V(0) < (r^*)^2$. The DCT bootstrap (`ContinuumInstability.lean`) proves $r(t) \ge r_{\min} > 0$ uniformly, which then enters the basin machinery of Theorem A.

Theorem B: Lorentzian (`lorentzian_ode_global_stability_complete`). Fully formalized, 0 sorry, 0 axioms. Global stability for the Bernoulli ODE with explicit solution formula. Finite-dimensional, no measure theory needed.

Theorem C: n-pole (`kuramoto_solved`). Fully formalized, 0 sorry, 0 axioms. Finite-dimensional, exponential decay rate $\mu = K c_{\min} \delta \delta^*$. Global stability for any $r(0) > 0$.

Theorem D: Complex OA + full PDE (`kuramoto_stability_universal`). Conditional, 0 sorry, 2 axioms. The complex OA convergence relies on nonlinear Landau damping [9] (axiom). The PDE-to-OA bridge relies on OA manifold attractivity [2] (axiom). Both axioms have opaque (non-vacuous) conditions encoded as `opaque Prop` types.

7.2 Theorem table

Name	Model	Hypotheses	Conclusion	S
`real_scalar_complete`	Scalar dissipative	$\int \gamma\, g < \infty$, $V(0) < (r^*)^2$	$r(t) \to r^*$	0
`real_scalar_global` (Thm. 1.4)	Scalar dissipative	$\int \gamma + \int 1/\gamma < \infty$, $K > K_c$	$r(t) \to r^*$	0
`lorentzian_ode_global_stability_complete`	Lorentzian ODE	$K > 2\gamma$, $r_0 \in (0, 1)$	$r(t) \to r^*$	0
`kuramoto_solved`	n-pole	$K > K_c$, $r(0) > 0$	$r(t) \to r^*$	0
`kuramoto_stability_universal`	Complex OA + PDE	Analytic g, $K > K_c$	$\|\eta\| \to r^*$	0

7.3 Significance of the scalar dissipative result

The critical coupling $K_c = 2/\int(1/\gamma)\,\mathrm{d}\mu$, the equilibrium profile $\alpha^*(\omega)$, and the order parameter r^* are *identical* between the scalar dissipative model (2) and the standard Kuramoto–OA system (1) (restricted to the positive real symmetric subspace). The global stability theorem (Theorem 1.4) therefore proves that the same positive real steady state is a global attractor for a scalar dissipative flow sharing the Kuramoto phase-transition threshold, the Kuramoto fixed point, and the Kuramoto self-consistency structure. The pair bound is a property of the self-consistency algebra (the equilibrium relation (4) and the integral constraint (3)), not the specific flow generating the transients. This makes the technique potentially transferable to other mean-field models with the same equilibrium structure.

7.4 Novelty

This work contributes four things that are new:

1. **Novel proof technique for the scalar dissipative model.** The pair bound (Proposition 2.2)—a pointwise algebraic inequality lifted to the continuum double integral via Fubini—has no precedent in the Kuramoto literature. Prior approaches to Kuramoto-type stability used Fourier analysis [1], Volterra equations [2], or spectral theory [6]. The pair bound applies to the scalar dissipative model (2); it does not directly apply to the standard complex OA (1).

2. **Machine-checked formalization.** To the best of our knowledge, this is the first Lean formalization of a stability theorem for a Kuramoto-type model. (We have not found competing formalizations in Coq, Isabelle, HOL Light, or Agda, but do not claim to have performed an exhaustive search.)

3. **Rotation cancellation (structural observation).** The rotation term $-i\omega$ in the complex OA cancels in the V derivative after equilibrium subtraction (Proposition 6.2). This is a structural observation relevant to the complex OA Lyapunov analysis; it does not by itself yield convergence (which requires external Landau damping input).

4. **Finite first moment coverage.** Many explicit OA reductions and fully closed finite-dimensional analyses rely on Lorentzian or rational distributions. The finite first moment condition $\int \gamma \, \mathrm{d}\mu < \infty$ (with $\gamma(\omega) \lesssim 1 + |\omega|$) covers Student-$t$ with $\nu > 1$ (including the infinite-variance case $1 < \nu \leq 2$), Gaussian, compact support, and mixtures.

7.5 Limitations

1. **Basin of attraction (resolved for the scalar dissipative model).** The basin stability result (Theorem 1.2) requires $V(0) < (r^*)^2$. This condition is discharged by Theorem 1.4 for the scalar dissipative model (2): the DCT bootstrap proves $r(t) \geq r_{\min} > 0$ for $K > K_c$, giving global stability without a basin condition. The restriction to the scalar dissipative model remains (not the standard complex OA (1)).

2. **Two cited axioms with opaque conditions.** Theorem D relies on two published results stated as Lean axioms with opaque (non-vacuous) conditions: (a) nonlinear Landau damping [9] for $V \to 0$, requiring linear stability, Sobolev smallness, and regularity of g; and (b) OA manifold attractivity [2] for the PDE-to-OA bridge. The conditions are encoded as `opaque Prop` types that cannot be trivially discharged. Formalizing the underlying spectral theory remains future work.

3. **Conclusion is $\mathrm{Re}(\eta)^2$, not $|\eta|^2$.** On the symmetric subspace $\eta \in \mathbb{R}$ (proved, 0 sorry), so $\mathrm{Re}(\eta)^2 = |\eta|^2$. But the Lean type states $\mathrm{Re}(\eta)^2 \to (r^*)^2$ rather than $|\eta|^2 \to (r^*)^2$.

4. **Scalar model scope.** The scalar dissipative model (2) is a separate dynamical system from the standard complex OA (1).

7.6 Open problems

1. ~~Remove the $V(0) < (r^*)^2$ basin condition.~~ **Resolved** for the scalar dissipative model: `r_stays_positive_supercritical` (0 sorry) proves $r(t) \geq r_{\min} > 0$ for $K > K_c$ via DCT bootstrap. Remains open for the standard complex OA equation (1) without the Landau damping axiom.

2. Formalize the nonlinear Landau damping axiom (currently citing [9]).
3. Formalize the OA manifold attractivity axiom (currently citing [2]).
4. Extend to distributions without finite first moment (Student-t with $\nu \leq 1$) beyond the Lorentzian special case.

Appendix: Reproducibility

Repository	`https://github.com/taejun-song/kuramoto-lean`
Commit	`e94219b`
Lean toolchain	`leanprover/lean4:v4.30.0-rc1`
Mathlib revision	`0c154d67` (Mathlib4)
Build command	`lake build`
Final theorems	`real_scalar_complete` (basin), `r_stays_positive_supercritical` (global $r > 0$), `hPsi_floor_of_r_liminf` (bridge), `lorentzian_ode_global_stability_complete`

Axiom check	`#print axioms real_scalar_complete`
	`#print axioms r_stays_positive_supercritical`
	`#print axioms hPsi_floor_of_r_liminf`
	`#print axioms lorentzian_ode_global_stability_complete`
	All output only `propext`, `Quot.sound`, `Classical.choice`

Sorry check	`grep -rn "^\s*sorry" KuramotoLean/` returns 0 matches in the stability chain files
Unrelated sorry	`NeuralMeanField.lean` (2), `MeanFieldLimit.lean` (4) — outside the dependency closure of any final theorem above

The final theorems transitively depend on no project axioms and no `sorry`. The unrelated sorry's in imported files are in theorems that do not appear in the dependency graph of any theorem listed above, verifiable via `#print axioms`.

References

[1] H. Dietert, Stability and bifurcation for the Kuramoto model, *J. Math. Pures Appl.* **105** (2016), 451–489.

[2] H. Dietert and B. Fernandez, The mathematics of asymptotic stability in the Kuramoto model, *Proc. R. Soc. A* **474** (2018), 20180467.

[3] H. Dietert, *Stability of the Kuramoto model*, PhD thesis, Université Paris Diderot, 2016.

[4] Y. Kuramoto, Self-entrainment of a population of coupled nonlinear oscillators, in *International Symposium on Mathematical Problems in Theoretical Physics*, Lecture Notes in Physics **39**, Springer, 1975, pp. 420–422.

[5] E. Ott and T. M. Antonsen, Low dimensional behavior of large systems of globally coupled oscillators, *Chaos* **18** (2008), 037113.

[6] H. Chiba, A proof of the Kuramoto conjecture for a bifurcation structure of the infinite-dimensional Kuramoto model, *Ergod. Th. & Dynam. Sys.* **35** (2015), 762–834.

[7] J. R. Engelbrecht and R. Mirollo, Is the Ott-Antonsen manifold attracting?, *Phys. Rev. Research* **2** (2020), 023057.

[8] S. H. Strogatz, From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators, *Physica D* **143** (2000), 1–20.

[9] H. Dietert, Stability of partially locked states in the Kuramoto model through Landau damping with Sobolev regularity, *arXiv:1707.03475* (2020).

[10] The Mathlib Community, *Mathlib4: A unified library of mathematics for Lean 4*, `https://leanprover-community.github.io/mathlib4_docs/` (2024).